

Outline

- › Physical structure
- › Current-voltage relationships
 - Regions (modes) of operation
- › MOSFET bias circuits (MOSFETs at dc)
- › Other topics
 - The body effect
 - The depletion-type MOSFET

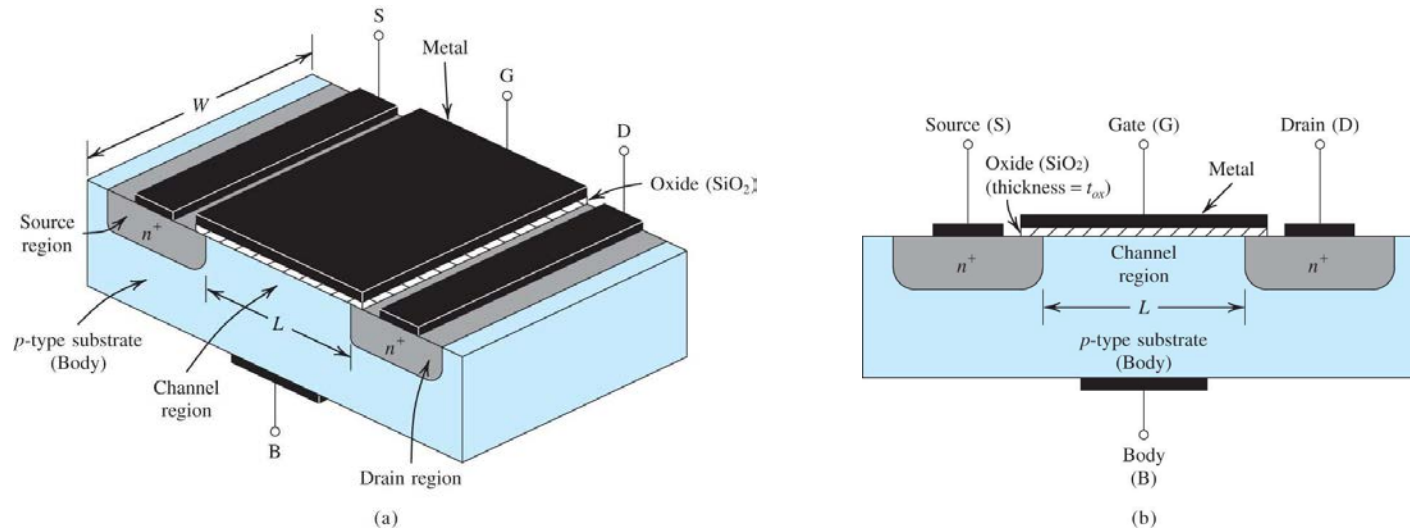
Introduction

- › So far we have looked at two-terminal semiconductor devices (e.g. diodes)
- › Three-terminal devices (i.e. transistors) are more useful because they have many applications
 - Amplifiers, switching circuits, digital logic devices etc.
- › Two major types of transistors:
 - Metal oxide semiconductor field effect transistors (**MOSFETs**)
 - Bipolar junction transistors (**BJTs**)

Some history

- › The transistor was developed at Bell labs around 1950
- › The first transistors were bipolar devices (BJTs)
- › Field effect devices (MOSFETs) were developed starting in the 1960s
- › Over time MOS-based semiconductor technologies surpassed bipolar technologies
- › Advancement in semiconductors led to large scale integrated devices such as microprocessor and memory chips
- › CMOS-based technologies dominate the field

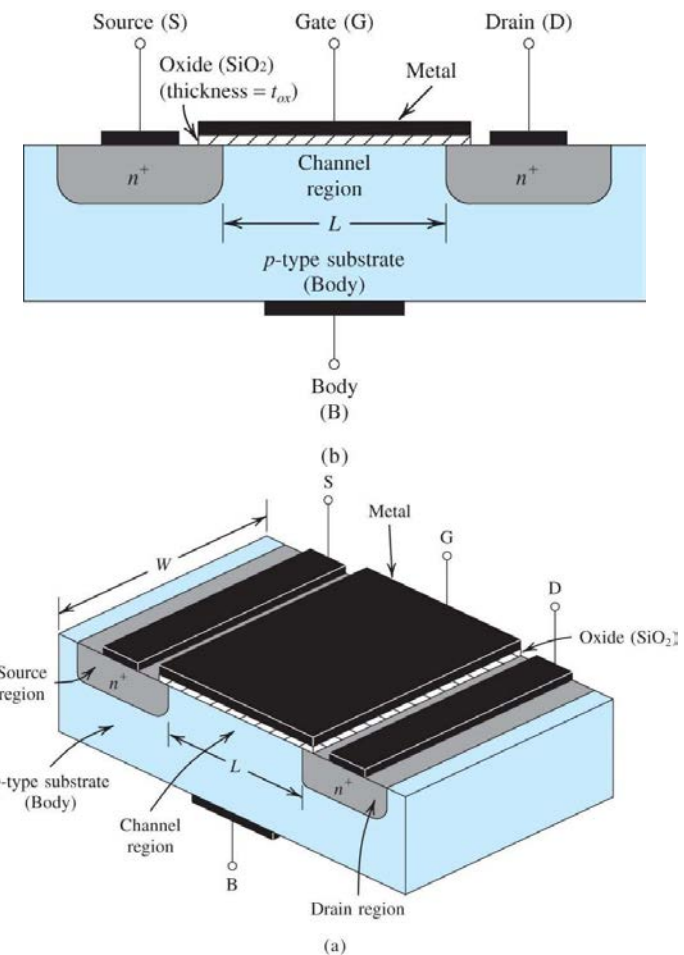
Physical Structure: n-channel MOSFET



Physical structure of the enhancement-type NMOS transistor: (a) perspective view; (b) cross section. Typically $L = 0.03 \mu\text{m}$ to $1 \mu\text{m}$, $W = 0.05 \mu\text{m}$ to $100 \mu\text{m}$, and the thickness of the oxide layer (t_{ox}) is in the range of 1 to 10 nm.

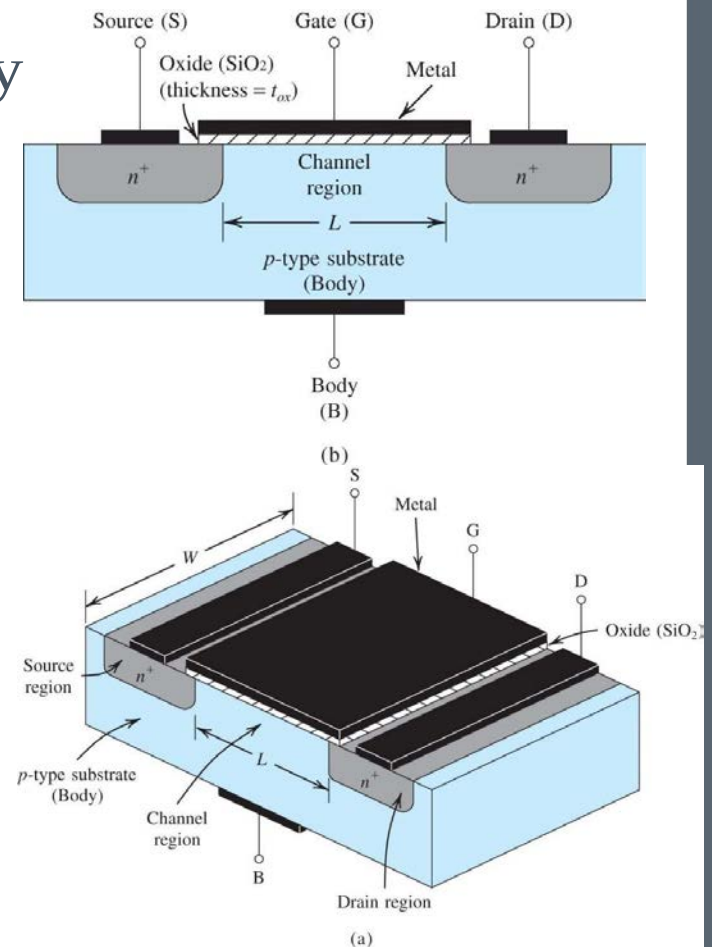
Physical structure: n-channel MOSFET

- › P-type substrate: a silicon wafer that provides support
- › Two doped n-type regions: **source** and **drain**
- › A thin layer of SiO_2 : an insulator
- › Metal placed on top of SiO_2 forms the **gate**
- › A **body** terminal tied to substrate
- › Thus, there are four terminals



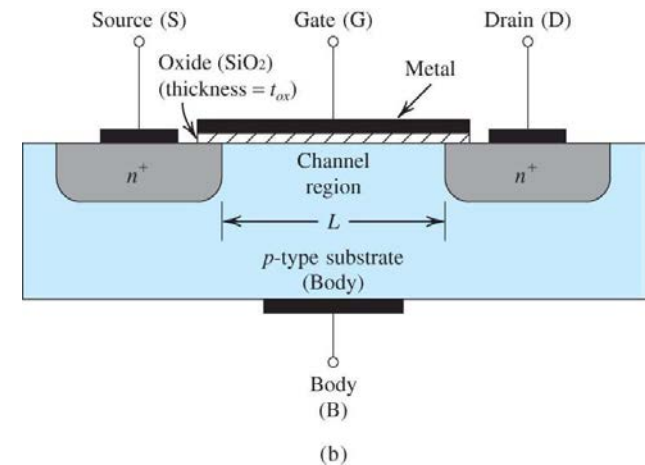
Physical structure: n-channel MOSFET

- › However, in most applications the body is connected to source
- › Three terminals: Gate, Drain, Source
- › There are two pn junctions in the device:
 - Substrate to source and substrate to drain
- › These two pn junctions are reverse-biased
- › Drain is at a higher potential than the source



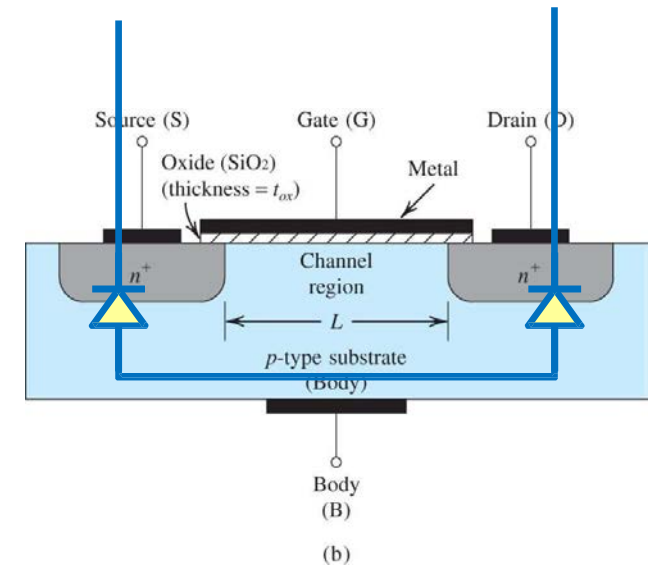
Operation with zero gate voltage

- › It will be shown that a voltage applied to the gate controls current flow between drain and source
- › Current will flow from drain to source through the channel
- › The channel region has length L and width W



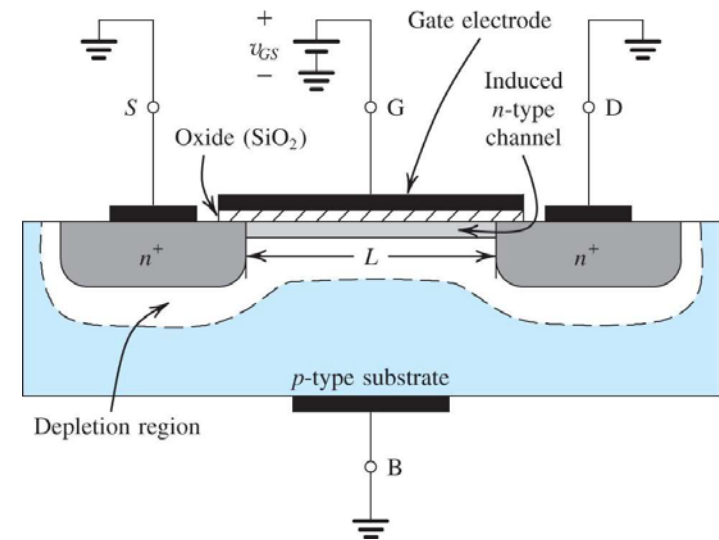
Operation with zero gate voltage

- › With zero voltage applied to gate, two back-to-back diodes exist in series between the drain and the source
 - One pn junction is between the p-type substrate and the n-type source
 - Another pn junction is between the p-type substrate and the n-type drain
- › The back-to-back diodes prevent current conduction from drain to source when a voltage v_{DS} is applied
- › Resistance between drain to source is very high (10^{12} ohm)



Creating a channel for current flow

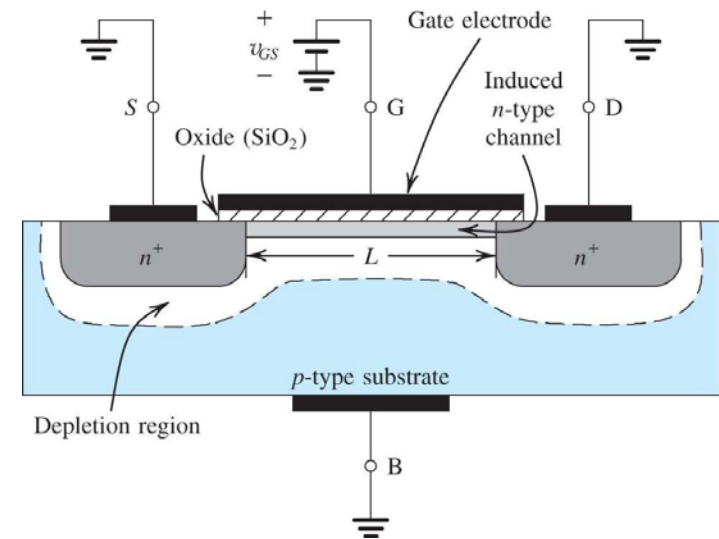
- › With the drain and source terminals grounded,
- › A gate to source voltage, v_{GS} is applied to the gate terminal, causing a build up of positive charge along the metal electrode
- › This build up causes free holes to be repelled from region of p-type substrate
- › These holes are pushed to the substrate leaving behind a depletion region



The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

Creating a channel for current flow

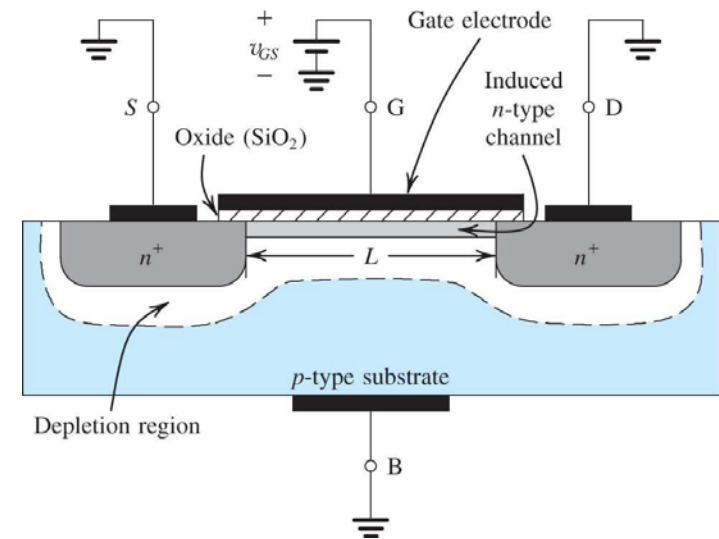
- › The positive gate voltage also attracts electrons from the n^+ source and drain regions into the channel
- › Once a sufficient number of these electrons accumulate, an n -region is created, connecting the source and drain regions
- › This provides a path for current flow between Drain and Source



The enhancement-type NMOS transistor with a positive voltage applied to the gate. **An n channel is induced at the top of the substrate beneath the gate.**

Creating a channel for current flow

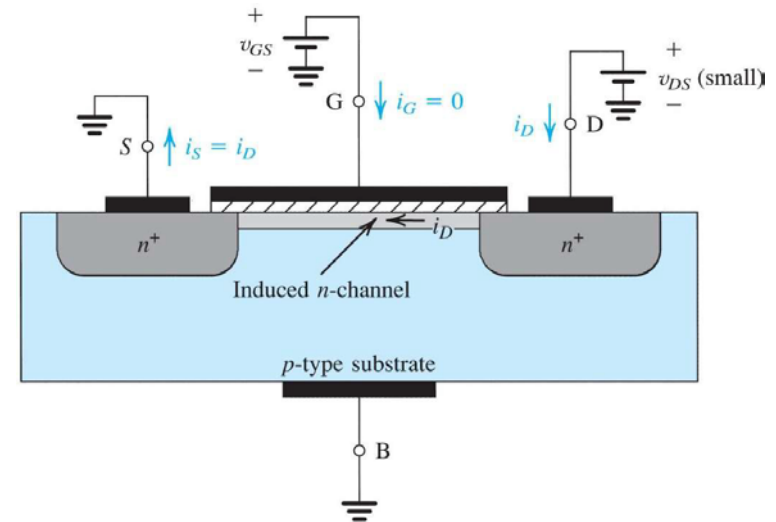
- › If a voltage is applied between drain and source, current flows through the induced n region
- › Thus, the induced n region forms a channel for current flow from drain to source
- › The MOSFET shown in the figure is known as an **n-channel** MOSFET or an **NMOS**



The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

Creating a channel for current flow

- › The minimum value of v_{GS} required to form a conducting channel between the drain and source is known as the **threshold voltage** V_t , typically between 0.3 and 1.0 Vdc (it can be higher)
- › When a positive v_{GS} is applied, an electric field develops between the gate electrode and the induced n-channel; this is known as the **field effect**



An NMOS with $V_{GS} > V_t$ and with a small v_{DS} applied

Creating a channel for current flow

- › The excess voltage over V_t is termed the overdrive voltage (in the textbook)

$$v_{OV} \equiv v_{GS} - V_t$$

- › The electron charge in the channel is expressed as

$$|Q| = C_{ox}(WL)v_{OV}$$

- › Where C_{ox} is the oxide capacitance (F/m²)
- › W and L are the width and length of the channel respectively
- › C_{ox} is given by

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

- › ϵ_{ox} is permittivity and t_{ox} is oxide thickness ($\epsilon_{ox}=3.45 \times 10^{-11}$ F/m for Si)

Applying a small v_{DS}

- › When a small voltage v_{DS} is applied a current i_D starts to flow through the induced n-channel
- › This current (i_D) is carried by free electrons traveling from source to drain
- › This drain current is defined in terms of charge per unit length of n-channel and the electron drift velocity

- › $i_D = (\text{charge per unit length})(\text{electron drift velocity})$

$$i_D = (C_{ox} W v_{OV}) \left(\frac{\mu_n v_{DS}}{L} \right) = q t = q \left(\frac{L}{v} \right) = q L (\text{vel})$$

$$i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} \right] v_{DS}$$

$\mu = \text{vel} / \vec{E}$
 $\text{vel} = \mu * \vec{E}$

- › For small values of v_{DS} , the n-channel acts like a variable resistance whose value is controlled by v_{OV}

$$\vec{E} = \frac{V_{DS}}{L}$$

Applying a small v_{DS}

- › Conductance is defined as

$$g_{DS} = \frac{i_D}{v_{DS}} = (\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV}$$

- › Conductance g_{DS} depends on:
 - $(\mu_n C_{ox})$ which is determined by the manufacturing process
 - $\left(\frac{W}{L} \right)$ or aspect ratio which depends on the geometry of the n-channel
 - $v_{OV} = v_{GS} - V_t$ which is applied externally
- › Process transconductance parameter is defined

$$k_n' = \mu_n C_{ox}$$

- › Multiplying by the aspect ratio yields the transconductance parameter

$$k_n = k_n' \frac{W}{L} = \mu_n C_{ox} \frac{W}{L}$$

Applying a small v_{DS}

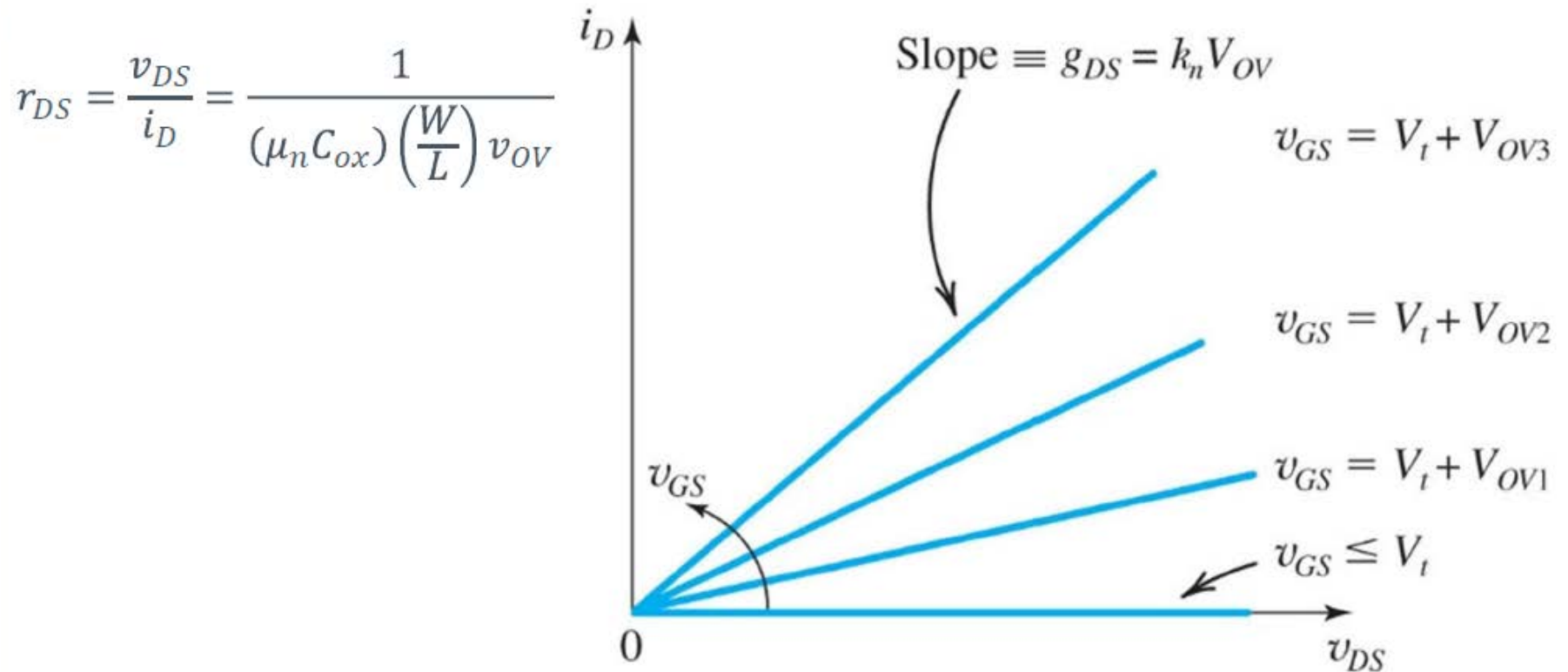
- › For small values of v_{DS} the MOSFET behaves as a linear resistance r_{DS}

$$r_{DS} = \frac{1}{g_{DS}} = \frac{v_{DS}}{i_D} = \frac{1}{(\mu_n C_{ox}) \left(\frac{W}{L}\right) v_{OV}}$$

- › r_{DS} depends on:
 - $(\mu_n C_{ox})$ which is determined by the manufacturing process
 - $\left(\frac{W}{L}\right)$ or aspect ratio which depends on the geometry of the n-channel
 - $v_{OV} = v_{GS} - V_t$ which is applied externally

Applying a small v_{DS}

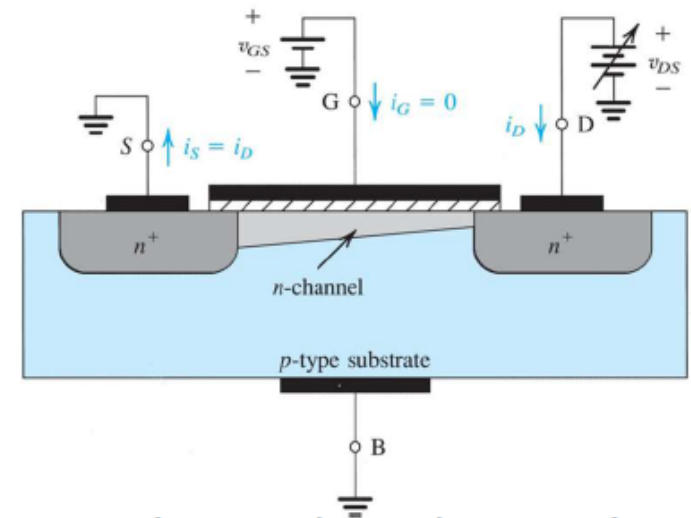
$k_n = \mu_n C_{ox} W/L = k'_n W/L$
is the transconductance parameter



The i_D - v_{DS} characteristics of the MOSFET when the voltage applied between drain and source V_{DS} is kept small

Operation as V_{DS} is increased

- › Let's consider the situation when V_{DS} is increased
- › V_{DS} is the voltage drop across the length of the channel
- › which means that the voltage across the channel (relative to the source) increases from 0 to V_{DS}
- › The voltage between the gate and points along the channel decreases from $V_{GS} = V_t + V_{OV}$ at the source to $v_{GD} = v_{GS} - v_{DS} = V_t + V_{OV} - v_{DS}$ at the drain
- › Since channel depth depends on voltage, the channel will no longer be of uniform depth
- › The channel is deepest at the source and shallowest at the drain (as shown)



Operation as v_{DS} is increased

- › The charge in the tapered channel is proportional to the channel area which is proportional to

$$\frac{1}{2} [V_{OV} + (V_{OV} - v_{DS})] = V_{OV} - \frac{1}{2} v_{DS}$$

- › From previous results

$$i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} \right] v_{DS}$$

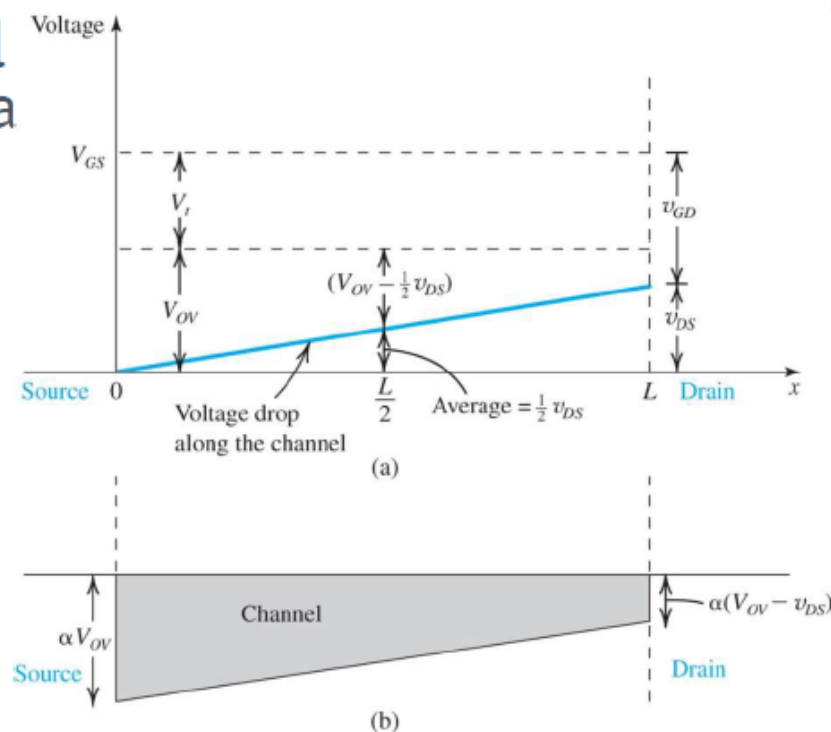
$$i_D = \left[k'_n \left(\frac{W}{L} \right) v_{OV} \right] v_{DS}$$

- › Substituting from above for V_{OV}

$$i_D = \left[k'_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} v_{DS} \right) \right] v_{DS}$$

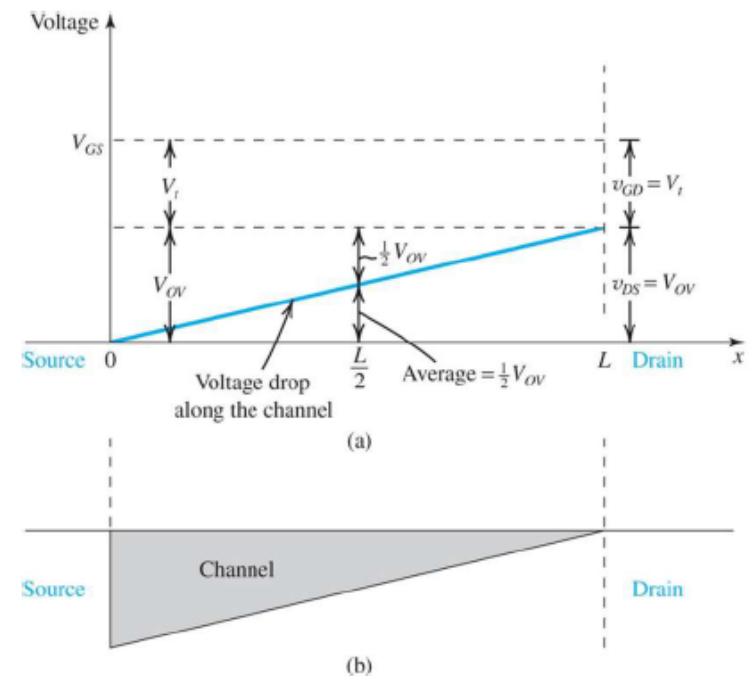
For a MOSFET with $v_{GS} = V_t + v_{OV}$ application of v_{DS} causes the voltage drop along the channel to vary linearly, with an average value of v_{DS} at the midpoint. Since $v_{GD} > V_t$ the channel still exists at the drain end. (b) The

Channel shape corresponding to the situation in (a) While the depth of the channel at the source is still Proportional to V_{OV} , the drain end is not.



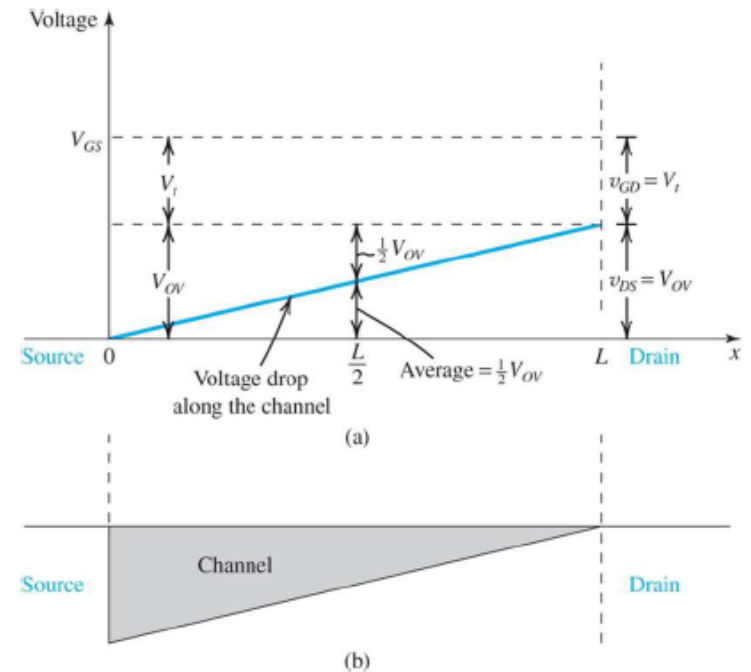
Operation as v_{DS} is increased

- $$i_D = \left[k'_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} v_{DS} \right) \right] v_{DS}$$
- $$i_D = \left[k'_n \left(\frac{W}{L} \right) \left(V_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right) \right]$$
- As v_{DS} is increased to v_{OV} , V_{GD} decreases to V_t
- and the channel length at the drain reduces to zero
- At this point, the MOSFET enters saturation mode of operation
- Further increasing v_{DS} (beyond v_{OV}) has no effect on the channel shape and i_D remains constant

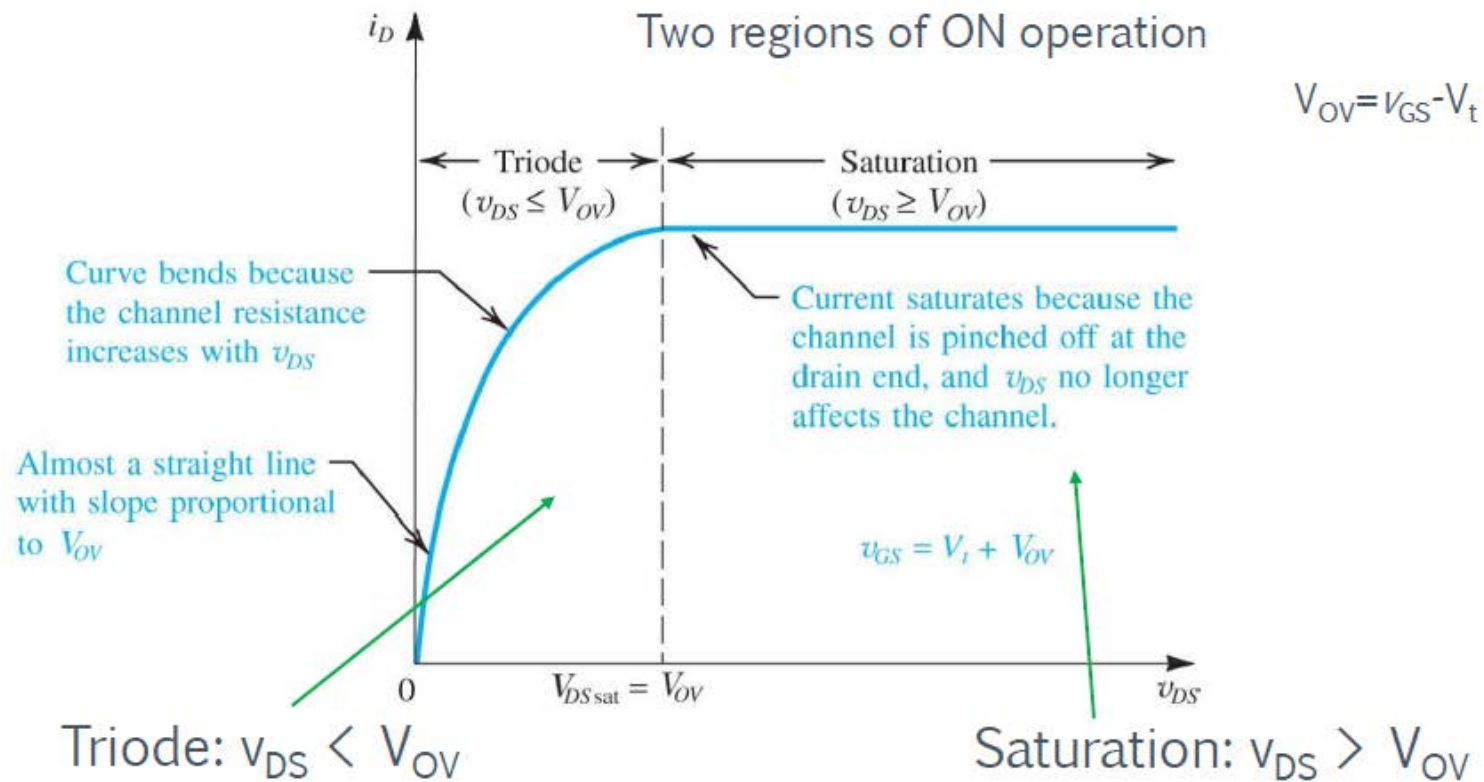


Operation as v_{DS} is increased

- › The drain current saturates at $v_{DS}=V_{OV}$
- › $i_D = \left[k'_n \left(\frac{W}{L} \right) \left(V_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right) \right]$
- › Substituting $v_{DS}=V_{OV}$ above
- › $i_D = \left[k'_n \left(\frac{W}{L} \right) \left(V_{OV} V_{OV} - \frac{1}{2} V_{OV}^2 \right) \right]$
- › $i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2$



i_D versus v_{DS} for an NMOS



$$i_D = \left[k'_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} v_{DS} \right) \right] v_{DS}$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2$$

Ex

Consider a process technology for which $L_{\min} = 0.4 \mu\text{m}$, $t_{\text{ox}} = 8 \text{ nm}$, $\mu_n = 450 \text{ cm}^2/\text{V} \cdot \text{s}$, and $V_t = 0.7 \text{ V}$.

- (a) Find C_{ox} and k'_n .
- (b) For a MOSFET with $W/L = 8 \mu\text{m}/0.8 \mu\text{m}$, calculate the values of V_{OV} , V_{GS} , and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu\text{A}$.
- (c) For the device in (b), find the values of V_{OV} and V_{GS} required to cause the device to operate as a $1000\text{-}\Omega$ resistor for very small v_{DS} .

(a)

$$\begin{aligned}C_{ox} &= \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{8 \times 10^{-9}} = 4.32 \times 10^{-3} \text{ F/m}^2 \\&= 4.32 \text{ fF}/\mu\text{m}^2 \\k'_n &= \mu_n C_{ox} = 450 \text{ (cm}^2/\text{V} \cdot \text{s)} \times 4.32 \text{ (fF}/\mu\text{m}^2) \\&= 450 \times 10^8 \text{ (}\mu\text{m}^2/\text{V} \cdot \text{s)} \times 4.32 \times 10^{-15} \text{ (F}/\mu\text{m}^2) \\&= 194 \times 10^{-6} \text{ (F/V} \cdot \text{s)} \\&= 194 \mu\text{A/V}^2\end{aligned}$$

(b) For operation in the saturation region,

$$i_D = \frac{1}{2} k'_n \frac{W}{L} V_{ov}^2$$

Thus,

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} V_{ov}^2$$

which results in

$$V_{ov} = 0.32 \text{ V}$$

Thus,

$$V_{GS} = V_t + V_{ov} = 1.02 \text{ V}$$

and

$$V_{DSmin} = V_{ov} = 0.32 \text{ V}$$

(c) For the MOSFET in the triode region with v_{DS} very small,

$$r_{DS} = \frac{1}{k'_n \frac{W}{L} V_{ov}}$$

Thus

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 \times V_{ov}}$$

which yields

$$V_{ov} = 0.52 \text{ V}$$

Thus,

$$V_{GS} = 1.22 \text{ V}$$